

## MSTAR PRACTICE WORKSHEET

## Lesson 9-3: Write Equations to Represent Relationships Between Two Variables

Grade 6 Mathematics · Standard 6.AT.11

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

### Part A — Skills Check

One point each.

**Question 1** SELECTED RESPONSE — CORRECT: C

Recreation World charges \$24.95 per month. What is the total cost of 2 months of membership?

- ☐ A \$24.95
- ☐ B \$26.95
- ☒ C \$49.90
- ☐ D \$74.85

Two months at \$24.95 each is  $2 \times 24.95 = 49.90$ .

**Question 2** SELECTED RESPONSE — CORRECT: A

In the equation  $c = 24.95m$  for the gym membership, what does  $m$  represent?

- ☒ A The number of months of membership
- ☐ B The total cost in dollars
- ☐ C The monthly fee of \$24.95
- ☐ D The name of the gym

$m$  stands for the quantity Miguel chooses — the number of months. The letter  $c$  stands for the total cost that results.

**Question 3**    **SELECTED RESPONSE — CORRECT: A**

A canoe rents for \$15 per hour. Let  $h$  = hours and  $c$  = total cost. Which equation represents the relationship?

- ☒ **A**  $c = 15h$
- ☐ **B**  $c = 15 + h$
- ☐ **C**  $h = 15c$
- ☐ **D**  $c = h - 15$

Each hour costs \$15, so the total is 15 times the number of hours:  $c = 15h$ .

**Question 4**    **SELECTED RESPONSE — CORRECT: C**

Using  $c = 15h$ , what is the total cost of a 3-hour canoe rental?

- ☐ **A** \$15
- ☐ **B** \$18
- ☒ **C** \$45
- ☐ **D** \$75

Substitute  $h = 3$ :  $c = 15 \times 3 = 45$ .

**Question 5**    **SELECTED RESPONSE — CORRECT: B**

In the gym membership situation, which quantity is the DEPENDENT variable?

- ☐ **A** The number of months
- ☒ **B** The total cost
- ☐ **C** The monthly fee of \$24.95
- ☐ **D** The number of gyms in town

The total cost depends on the number of months of membership — Miguel chooses the months, and the cost responds.

**Question 6**

SELECTED RESPONSE — CORRECT: B

The membership table shows: 1 month, \$24.95; 2 months, \$49.90; 3 months, \$74.85. By how much does the total cost increase each month?

- ☐ A \$24.05
- ☒ B \$24.95
- ☐ C \$25.00
- ☐ D \$49.90

Each row grows by the monthly fee:  $49.90 - 24.95 = 24.95$ , and  $74.85 - 49.90 = 24.95$ .

**Part B — Test-Format Questions**

Scoring notes and distractor rationales below each item.

**Question 7****TWO-PART QUESTION****Part A — correct answer: D**

A spin studio charges \$7.50 for each class. Let  $n$  be the number of classes a member takes and let  $c$  be the total cost in dollars. Which equation represents the relationship?

- (A)  $c = 7.5 + n$
- (B)  $n = 7.5c$
- (C)  $c = n \div 7.5$
- (D)  $c = 7.5n$

The studio charges \$7.50 for every class, so the total cost is the rate times the number of classes:  $c = 7.5n$ . The dependent variable, the cost, stands alone on the left, and the independent variable, the classes, is multiplied by the rate.

- **A:** This adds the price once rather than charging it per class. Each of the  $n$  classes costs \$7.50.
- **B:** This makes the number of classes depend on the cost. The cost is what depends on the classes.
- **C:** This divides by the price. Each class ADDS \$7.50, so multiply.

**Part B — correct answer: D**

Which reasoning justifies the equation chosen in Part A?

- (A) The number of classes is added to the price of one class, so 4 classes cost  $7.5 + 4 = \$11.50$ .
- (B) The number of classes depends on the cost, so the classes belong alone on the left side of the equation.
- (C) The \$7.50 is shared among the classes, so 4 classes cost  $7.5 \div 4 = \$1.875$  each.
- (D) The \$7.50 is charged for every single class, so the total is 7.5 times the classes; 4 classes cost  $7.5 \times 4 = \$30$ .

A rate applies to every class, so the operation is multiplication and the total for 4 classes is  $7.5 \times 4 = \$30$ . Adding gives \$11.50, which charges the fee only once. The cost depends on the classes, not the other way around, so the cost is the variable that stands alone.

**Question 8****SELECT ALL THAT APPLY — CORRECT: A, C, E**

A community pool charges \$9 for each hour of lane rental. Let  $h$  be the number of hours rented and let  $c$  be the total cost in dollars. Select ALL statements that correctly represent this relationship.

**A**  $c = 9h$

**B**  $h = 9c$

**C** The total cost is 9 times the number of hours rented.

**D**  $c = 9 + h$

**E** In the equation,  $h$  stands for the number of hours rented and  $c$  stands for the total cost in dollars.

**F** In the equation,  $c$  stands for the \$9 charged for each hour.

The \$9 is charged for every hour, so the total cost is the rate times the hours:  $c = 9h$ , which also checks against 4 hours at  $9 \times 4 = \$36$ . Writing  $h = 9c$  makes the hours depend on the cost, and  $c = 9 + h$  charges the fee only once. Defining the variables means saying what each letter counts:  $h$  counts hours and  $c$  counts dollars of total cost, while the 9 is the fixed rate and is not a variable.

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Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.